Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18SE5DE11/18IOT5DE11]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

CLOUD COMPUTING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	As ICICI Lombard was moving towards a more customer-centric approach with cutting edge technology, there was one unsolved mystery to be tackled, 'How to handle applications with peak times that are exponentially higher than the average?' Provisioning and de-provisioning of servers based on these varying peak loads caused time, cost and resource pressures on the organization.	**07**
	b.	A group of college freshers have a well-thought business idea. They want to start an SMB (Small Medium-sized Business) at a budget (preferably on a pay-as-you-go basis) but with as much less overhead of maintaining the infrastructure as possible. Which cloud deployment model would be suitable here and why? Compare and contrast public cloud and private cloud deployment models.	**07**
2.	**a.**	The Jain university wants to provide education services such as study material, video and virtual labs for the education community by connecting IIT, NIT and reputed private universities in India. Which cloud deployment model will suit this need? As cloud engineers justify your answers.	**07**
	b.	Compare the three cloud computing delivery models, SaaS, PaaS, and IaaS, from the point of view of application developers and users. Analyze the differences between PaaS and IaaS.	**07**
3.	**a.**	Elaborate on the usage of the AWS EC2. Write the steps to be followed to create an EC2 instance and connect the virtual machine using SSH protocol.	**07**
	b.	SAGO is a financial company that wants to move its legacy applications into the cloud environment. Write efficient seven steps for migrating to the cloud.	**07**

OR

4.	**a.**	Ficuslot Innovation Pvt Ltd is a website development company that got web development projects from customers. 4. Ficuslot Innovation Pvt Ltd company needs a virtual environment where they can build and deploy the web development project in the cloud and wants to provide service to their customers. Which cloud service model is suitable for this requirement? Justify your answer with proper explanation?	**07**
	b.	Sketch the OpenStack architecture and discuss the components of open stack architectures.	**07**
5.	**a.**	Describe how Raid technologies can be applied to improve the performances of the cloud storage system. Which RAID level would you recommend for the software industry?	**07**

	b.	Identify the main security threats for the SaaS cloud delivery model on a public cloud. Discuss the different aspects of these threats on a public cloud vis-à-vis the threats posed to similar services provided by a traditional service-oriented architecture running on private infrastructure.	07

OR

6.	a.	Analyze the reasons for the introduction of Network-attached storage (NAS) and their properties.	07
	b.	Discuss seven step model of Migration into a cloud.	07
7.	a.	Delhi public school wants to implement virtual desktop infrastructure instead of the traditional desktop environment. Sketch the VDI architectures and explain the advantage of using VDI instead of using traditional desktop environments.	07
	b.	Explain the benefits of virtualization. Compare the traditional computations and virtualized computations.	07

OR

8.	a.	Is there any way to make a cost comparison of the application that you have hosted on-premises and the application that's hosted in your Cloud environment? Compare the cost on-premises and the application that's hosted in your Cloud environment for the following configurations. Number Virtual machines-3 CPU Cores -4 RAM-8 GB Guest -OS-Linux Storage SAN-1 TB	07
	b.	Demonstrate the use of Service-oriented architecture. How it contributes to the cloud service delivery model.	07